bgsub vs pyREyes vs superxds / XDS on real frames from every binning and all three detectors (compare v1, 2026-11-05)

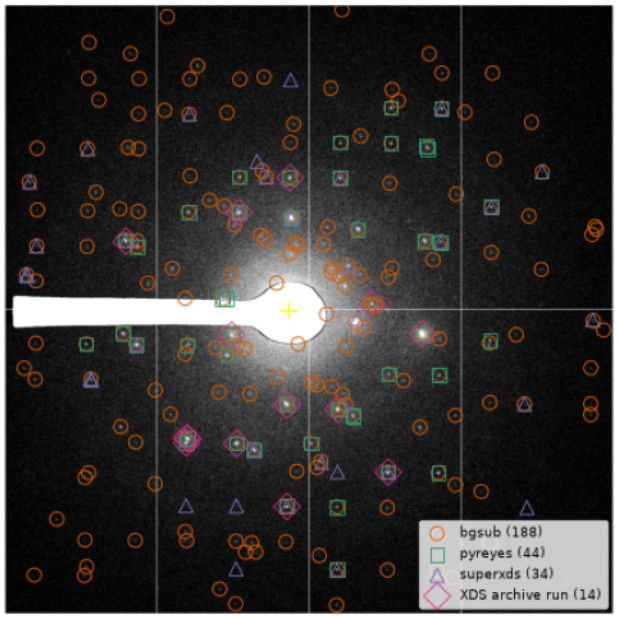
One dataset per detector x binning; the same three methods per frame. Peaks per frame = median over 5 frames [min-max]; peaks within 3 px of the mask are dropped for every method. superxds = the numpy port of the superxds / XDS math, validated against XDS (defaults BACKGROUND_PIXEL 2, SIGNAL_PIXEL 3, 6 px); 'owner XDS run' = the archive's own XDS run with stricter thresholds. Pages 2-11: one full-frame panel and one central-crop panel per dataset.

dataset	counts/px	bgsub law	bgsub peaks	pyREyes peaks	superxds spots (defaults)	owner XDS r	bgsub phi	ms/frame: bgsub / pyREyes / COLSPOT
Apollo bin 4, 2048^2	1.63	apollo(4)+1	156 [132-188]	50 [36-54]	34 [0-56]	13	1.31	157 / 152 / 359
Apollo bin 8, 1024^2	0.08	apollo(8)	22 [16-48]	20 [9-25]	4 [0-6]	-	1.04	57 / 30 / 77
Apollo bin 16, 512^2	0.83	apollo(16)	53 [47-75]	19 [15-21]	2 [0-7]	-	1.45	182 / 8 / 18
CETA-D bin 2, 2048^2	58.66	poisson(g 3.5, rv 8) x f 2.8; XDS gain 4.12	101 [74-136]	20 [7-27]	12 [0-35]	16	0.60	1428 / 152 / 351
TVIPS bin 2, 2048^2	53.69	poisson(g 10.3, rv 0) x f 3.1; XDS gain 6.929	34 [22-40]	9 [4-9]	12 [0-16]	3	0.38	186 / 144 / 353

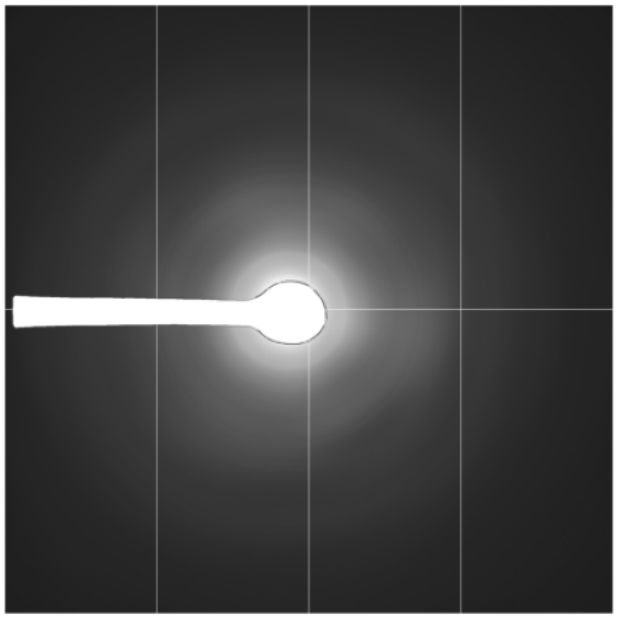
Reading. bgsub: a per-frame model of the background; its subtracted field is flat with the spots standing alone, weak spots included. superxds / XDS: a sweep-averaged table of 7 x 7 box means rescaled by one scalar per frame; grainy, with halo and ring residuals, finds the bright spots, fails on single sparse bin-8 frames. pyREyes: no background at all (difference of Gaussians + a global threshold), the brightest spots only.
CETA-D and TVIPS are integrating detectors with spatially correlated noise: bgsub ran with a read-noise + correlation law estimated from the frame (a 3 x 3 window factor of 2.8 / 3.1; declared, not calibrated); with a plain Poisson law it had flagged 8-47 % of the pixels. A calibrated law for these detectors is the open roadmap item. Both bgsub and superxds leave the glow crescent next to the CETA-D / TVIPS stop edge, wider than the penumbra model.
Full report: notes/reports/2026-11-05_compare_v1_images.md; figures notes/figures/compare_v1/; data ~/pod_data/bgsub/compare_v1/ (frames referenced, private copies).

Arctica-Apollo bin 4 (2048^2): rotation movie, 1 deg per frame, CL 420

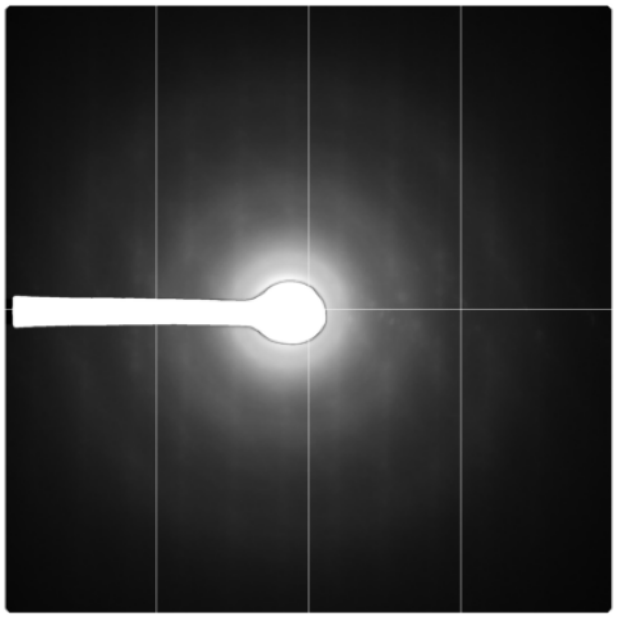
apollo_b4: original frame 49 (log), all methods' peaks



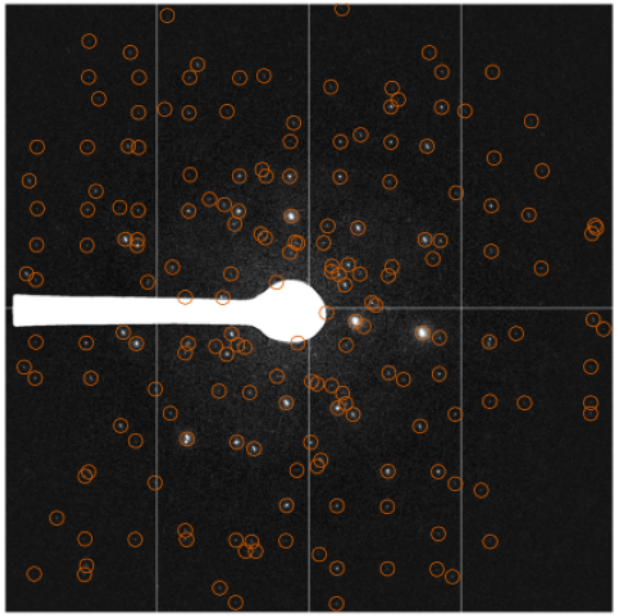
bgsub v2.1 background (log)



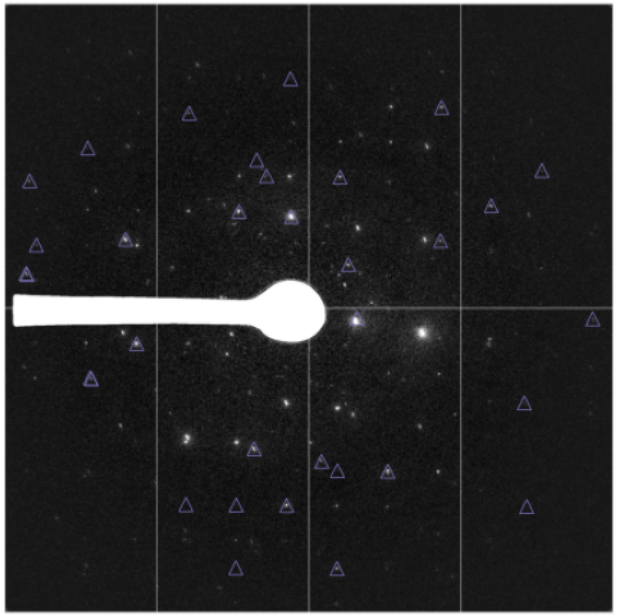
superxds / XDS background = BKGINIT x RATIO (log, same scale)



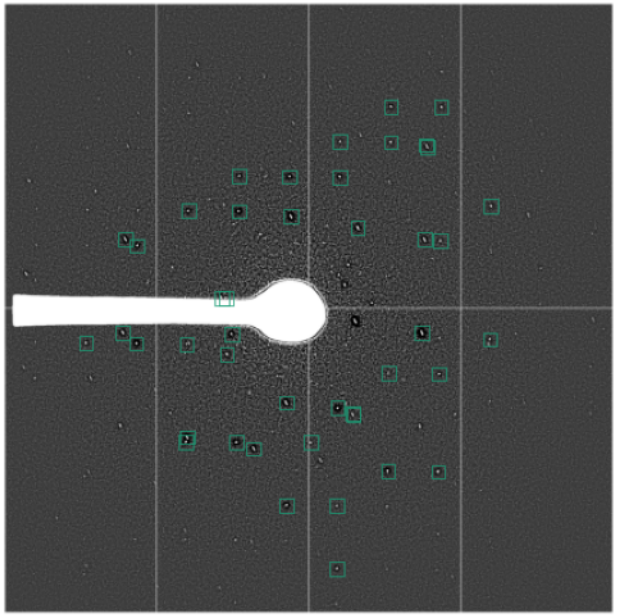
bgsub: frame - background, peaks (188)



superxds: frame - BKGINIT x RATIO, spots (34)

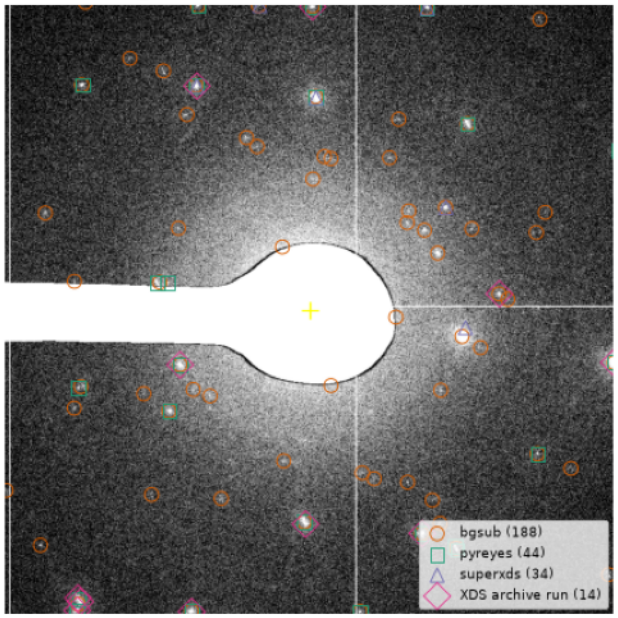


pyREyes: difference of Gaussians (no background), peaks (44)

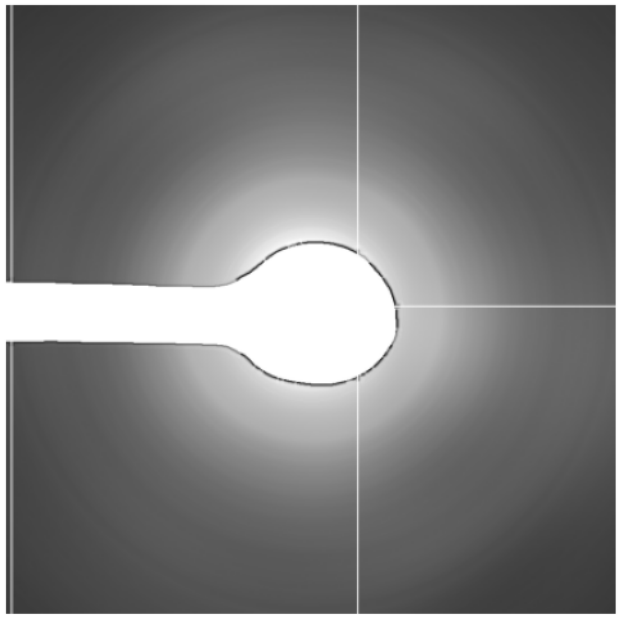


Arctica-Apollo bin 4 (2048^2): rotation movie, 1 deg per frame, CL 420 - central crop

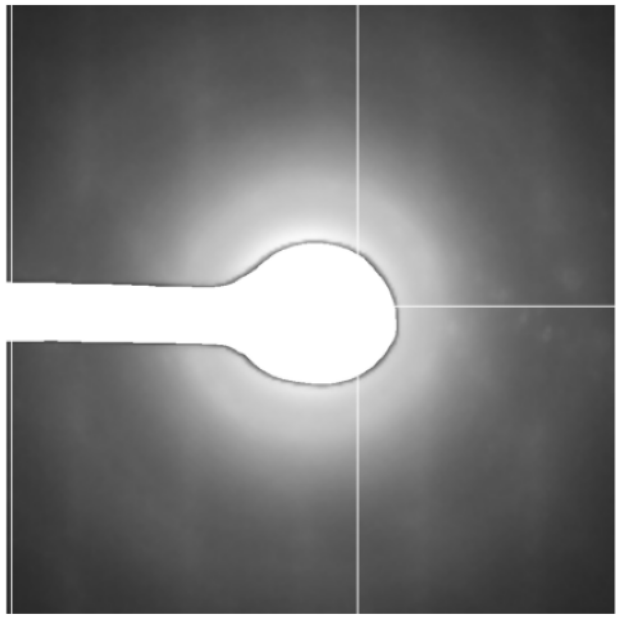
apollo_b4: original frame 49 (log), all methods' peaks



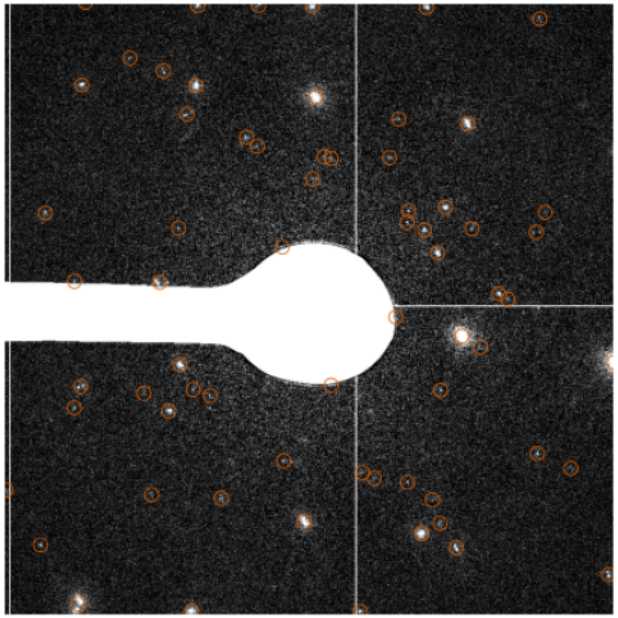
bgsub v2.1 background (log)



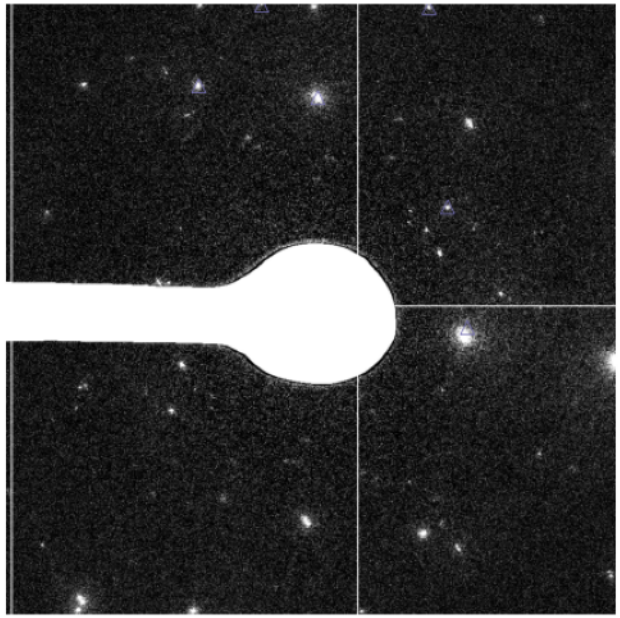
superxds / XDS background = BKGINIT x RATIO (log, same scale)



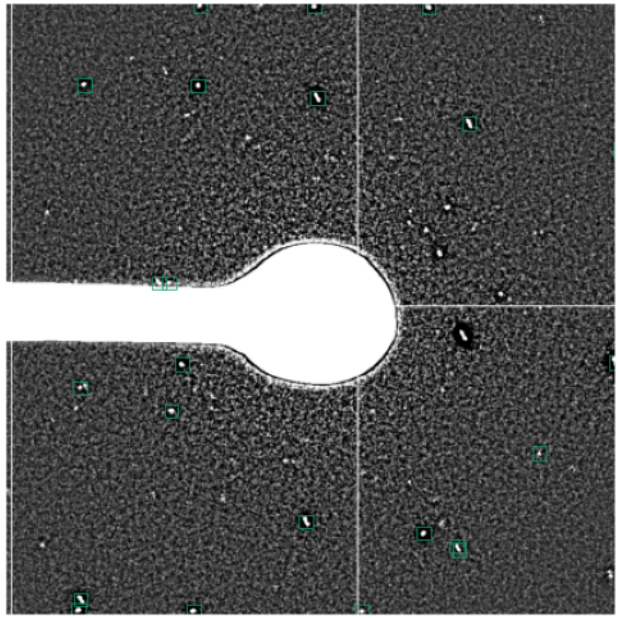
bgsub: frame - background, peaks (188)



superxds: frame - BKGINIT x RATIO, spots (34)

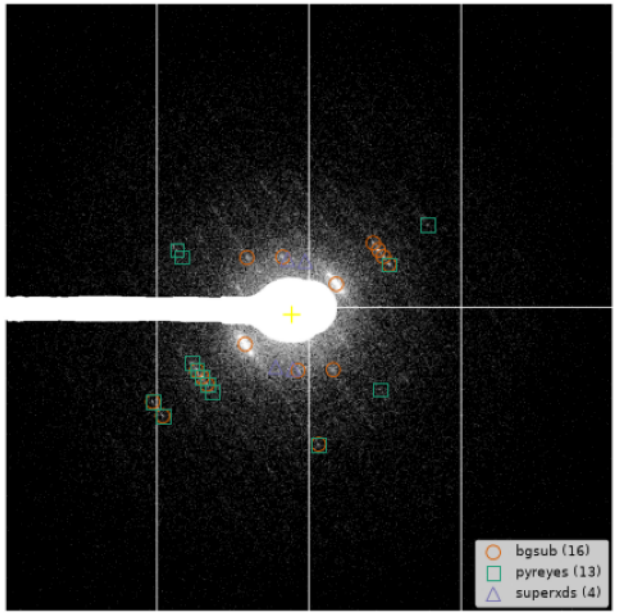


pyREyes: difference of Gaussians (no background), peaks (44)

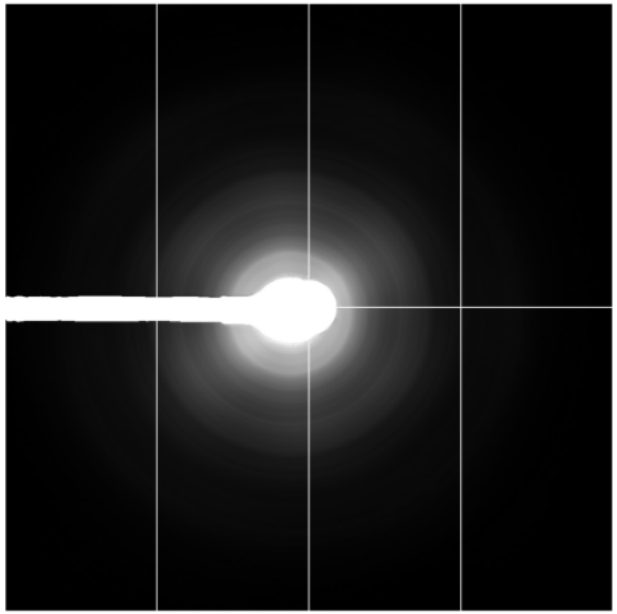


Arctica-Apollo bin 8 (1024^2): fly-scan Good snapshots, 49 ms, CL 420 (pseudo-sweep of 15)

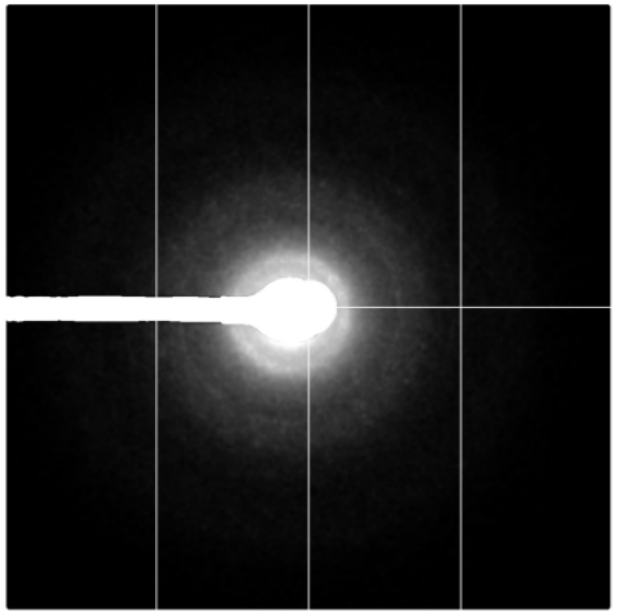
apollo_b8: original frame 3 (log), all methods' peaks



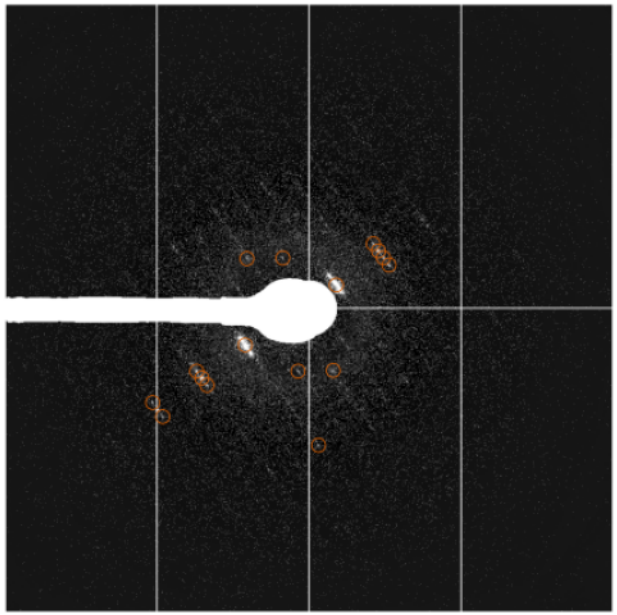
bgsub v2.1 background (log)



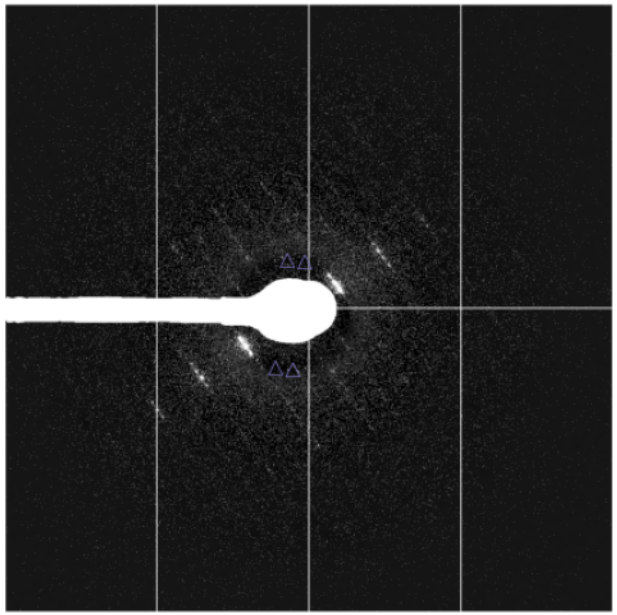
superxds / XDS background = BKGINIT x RATIO (log, same scale)



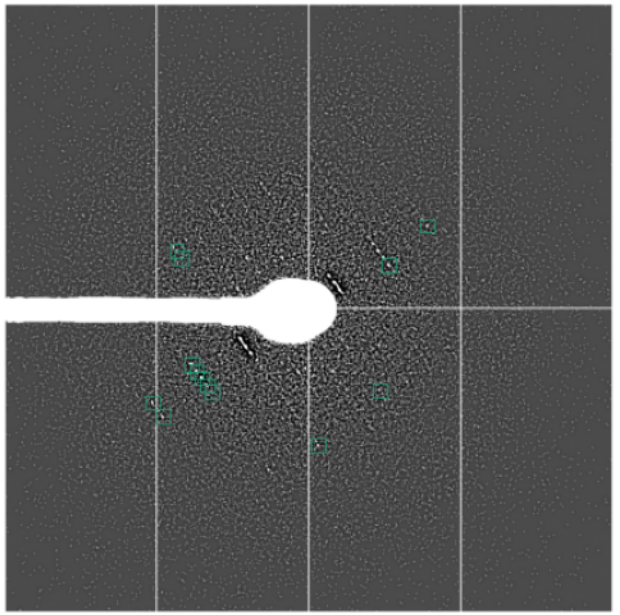
bgsub: frame - background, peaks (16)



superxds: frame - BKGINIT x RATIO, spots (4)

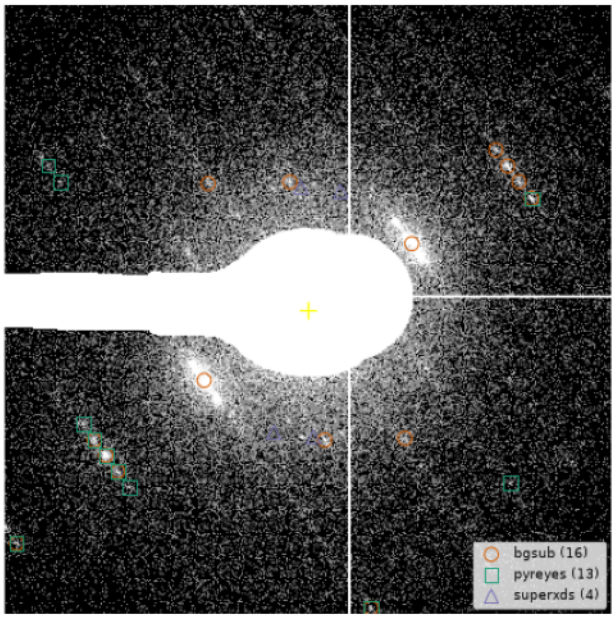


pyREyes: difference of Gaussians (no background), peaks (13)

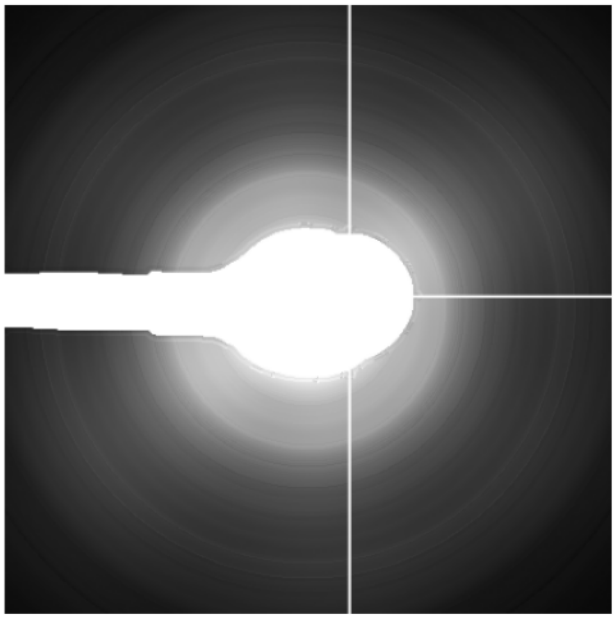


Arctica-Apollo bin 8 (1024^2): fly-scan Good snapshots, 49 ms, CL 420 (pseudo-sweep of 15) - central crop

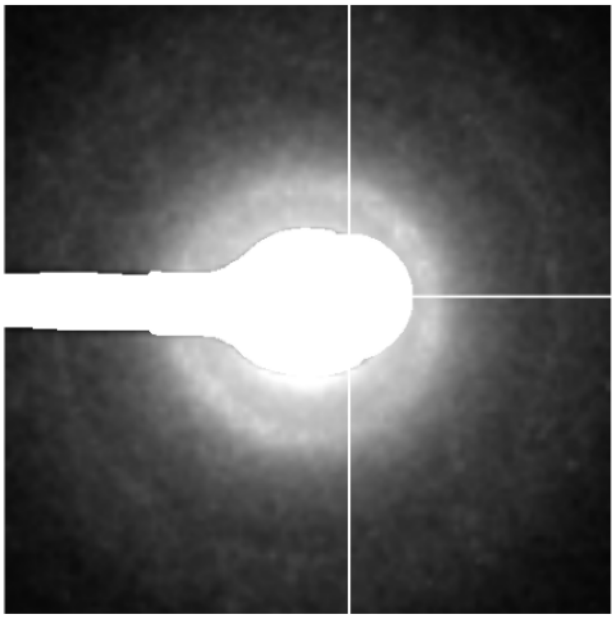
apollo_b8: original frame 3 (log), all methods' peaks



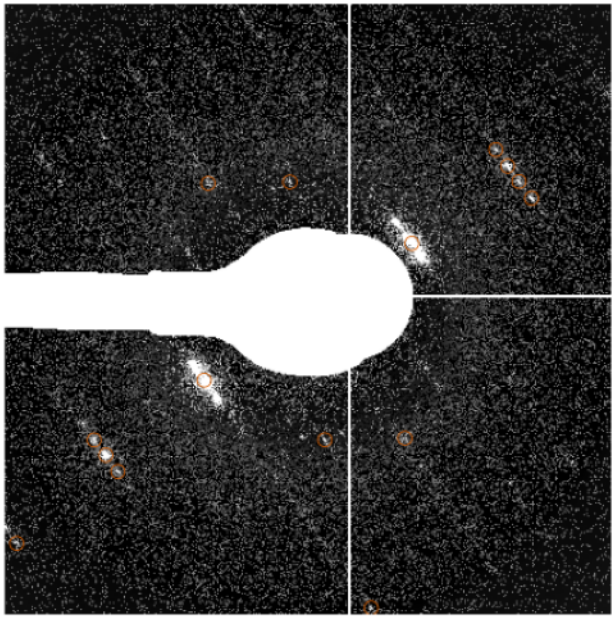
bgsub v2.1 background (log)



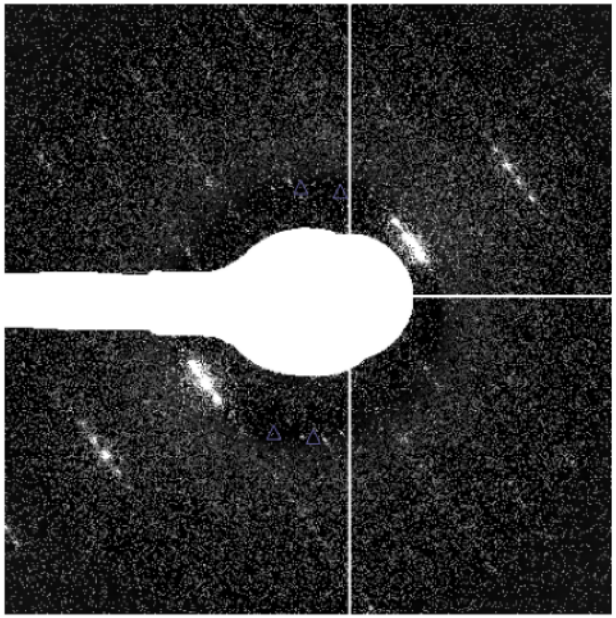
superxds / XDS background = BKGINIT x RATIO (log, same scale)



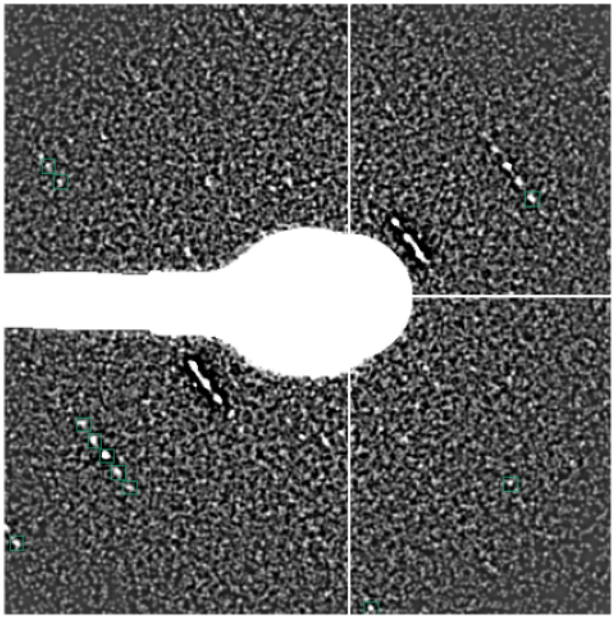
bgsub: frame - background, peaks (16)



superxds: frame - BKGINIT x RATIO, spots (4)

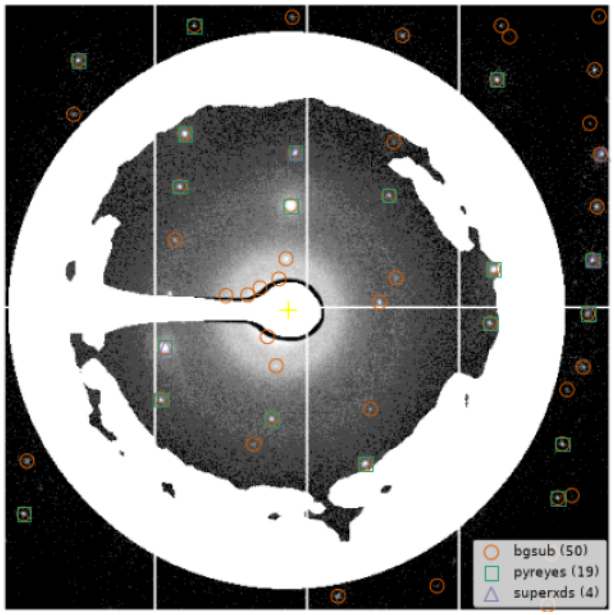


pyREyes: difference of Gaussians (no background), peaks (13)

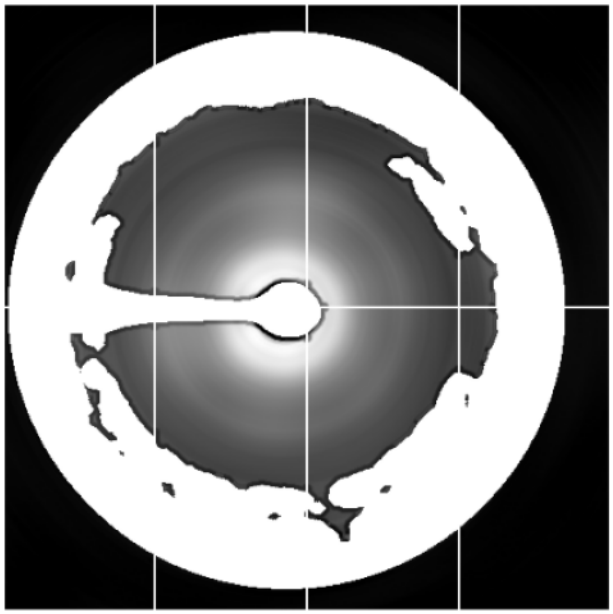


Arctica-Apollo bin 16 (512^2): fly-scan verify miniscan 2026-08-31, 0.196 s, CL 420, bin 16 (pseudo-sweep of 55)

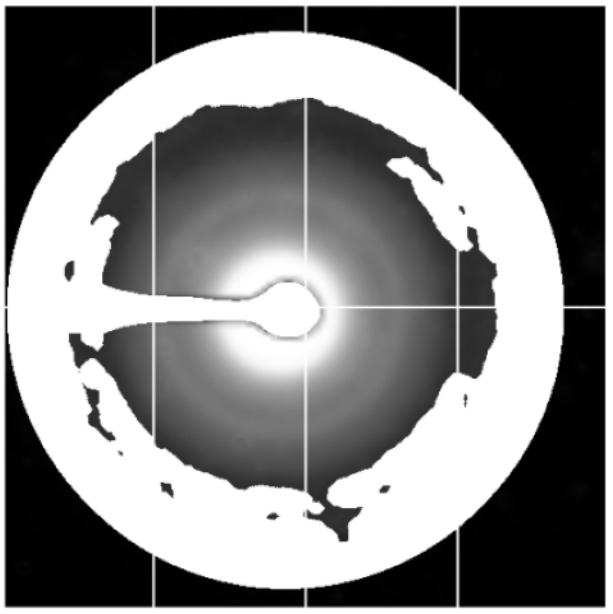
apollo_b16: original frame 28 (log), all methods' peaks



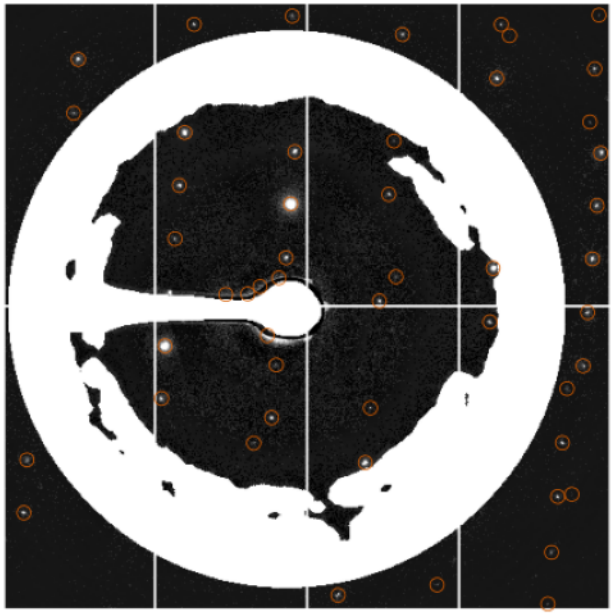
bgsub v2.1 background (log)



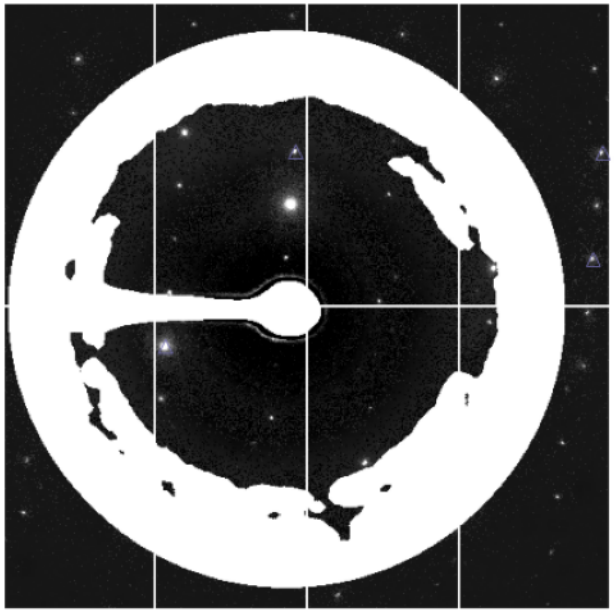
superxds / XDS background = BKGINIT x RATIO (log, same scale)



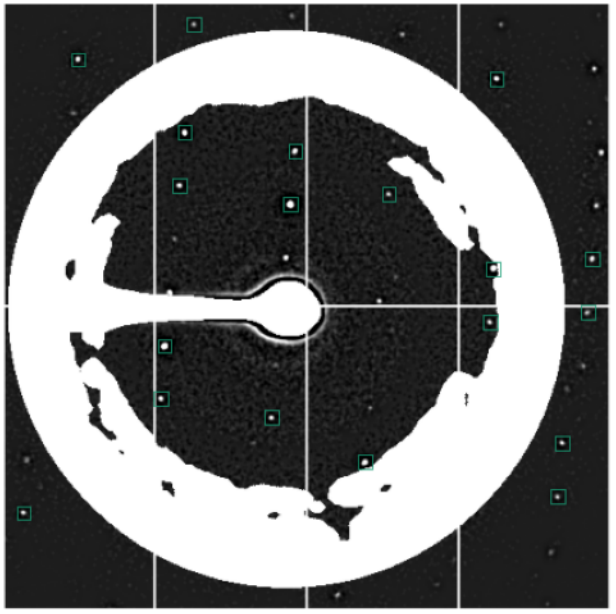
bgsub: frame - background, peaks (50)



superxds: frame - BKGINIT x RATIO, spots (4)

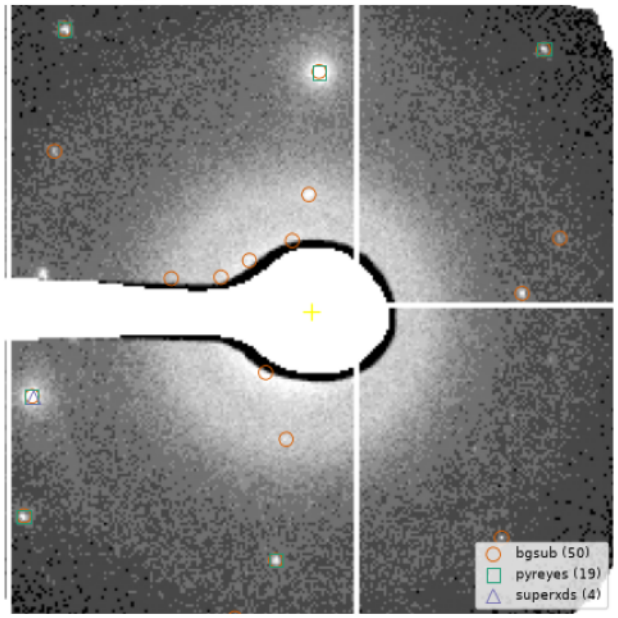


pyREyes: difference of Gaussians (no background), peaks (19)

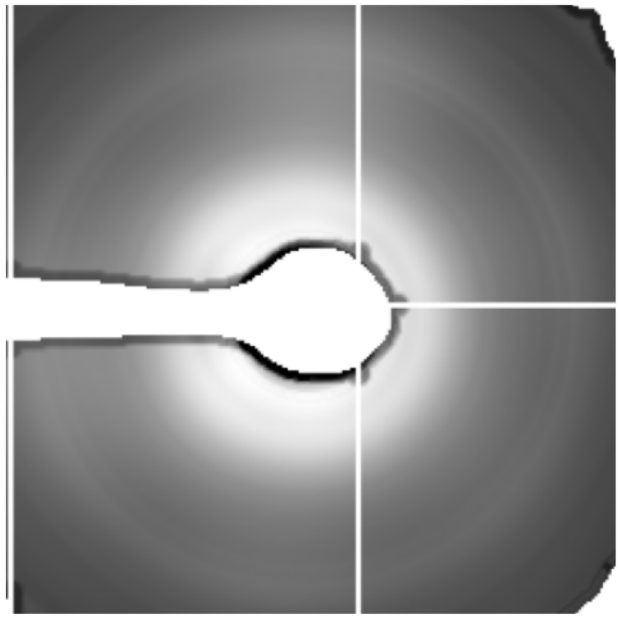


Arctica-Apollo bin 16 (512^2): fly-scan verify miniscan 2026-08-31, 0.196 s, CL 420, bin 16 (pseudo-sweep of 55) - central crop

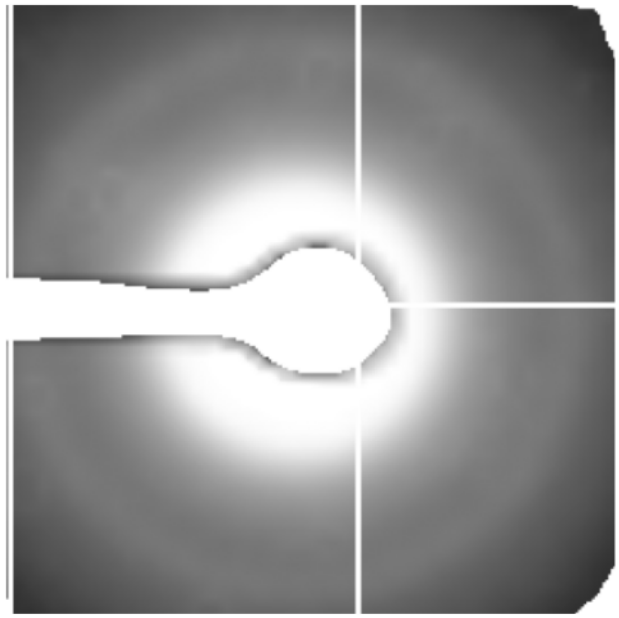
apollo_b16: original frame 28 (log), all methods' peaks



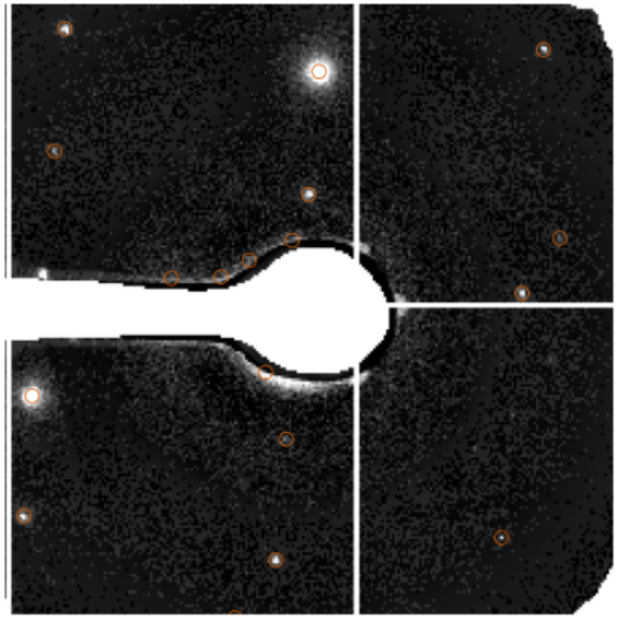
bgsub v2.1 background (log)



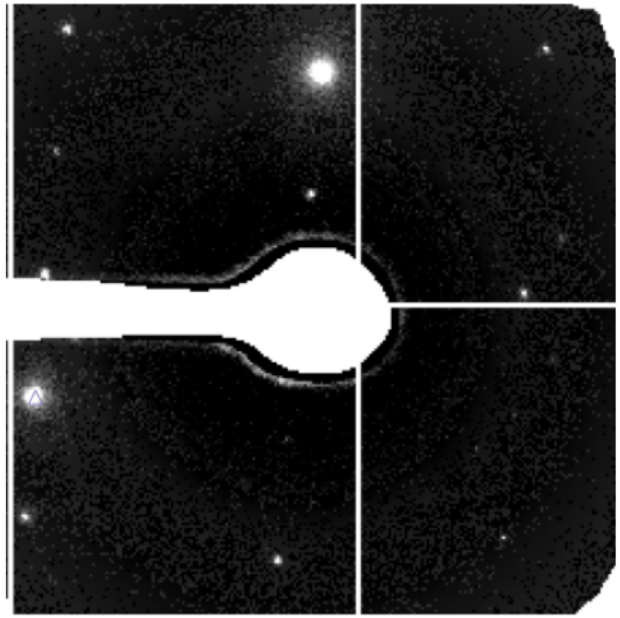
superxds / XDS background = BKGINIT x RATIO (log, same scale)



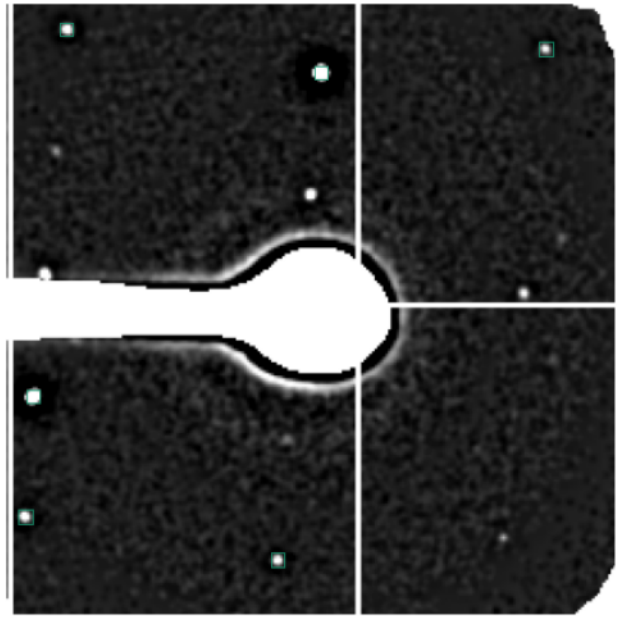
bgsub: frame - background, peaks (50)



superxds: frame - BKGINIT x RATIO, spots (4)

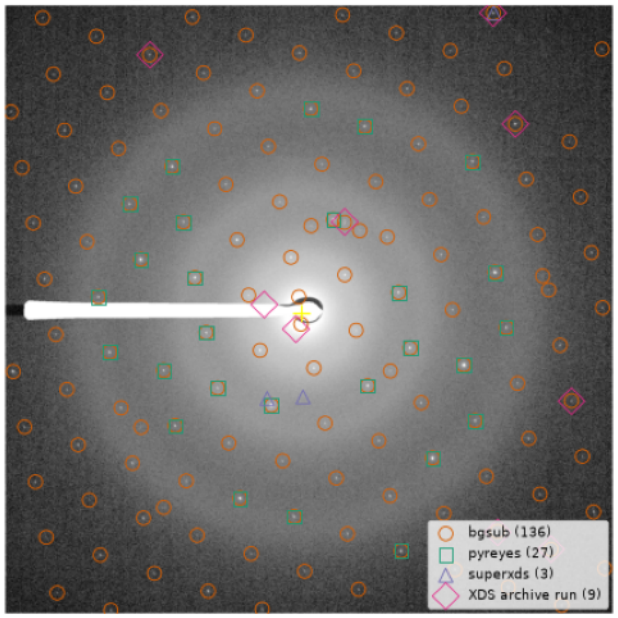


pyREyes: difference of Gaussians (no background), peaks (19)

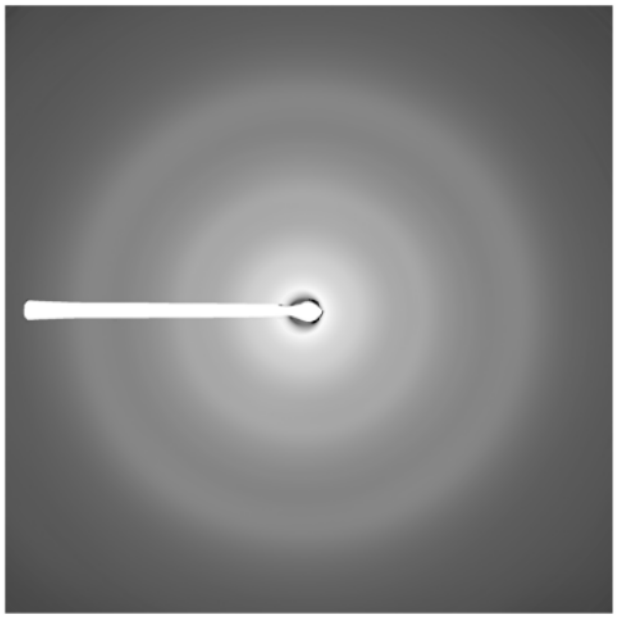


Arctica-CETA bin 2 (2048^2): rotation movie, 0.9 deg per frame, CL 960, CETA-D 4k binned 2

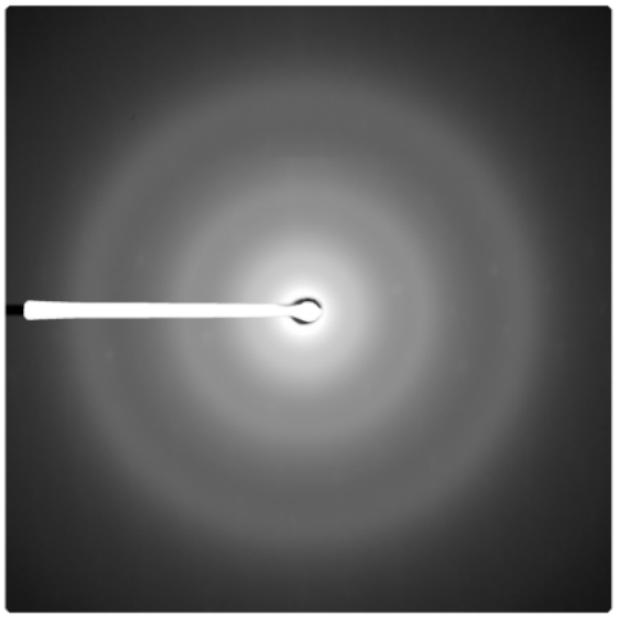
ceta_b2: original frame 56 (log), all methods' peaks



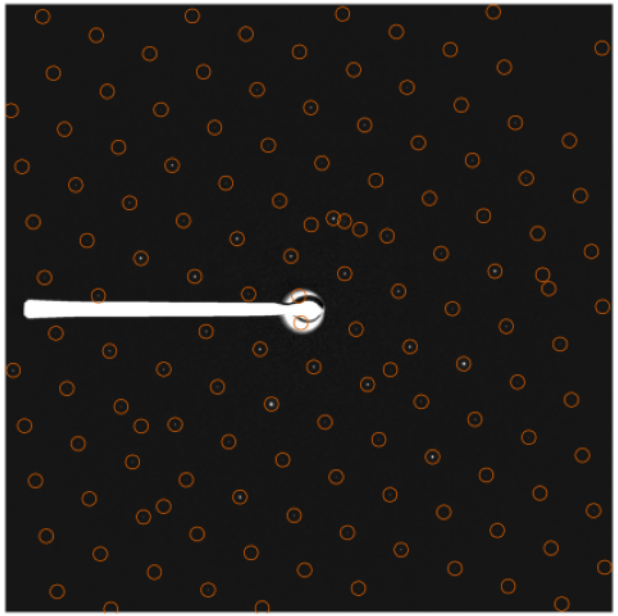
bgsub v2.1 background (log)



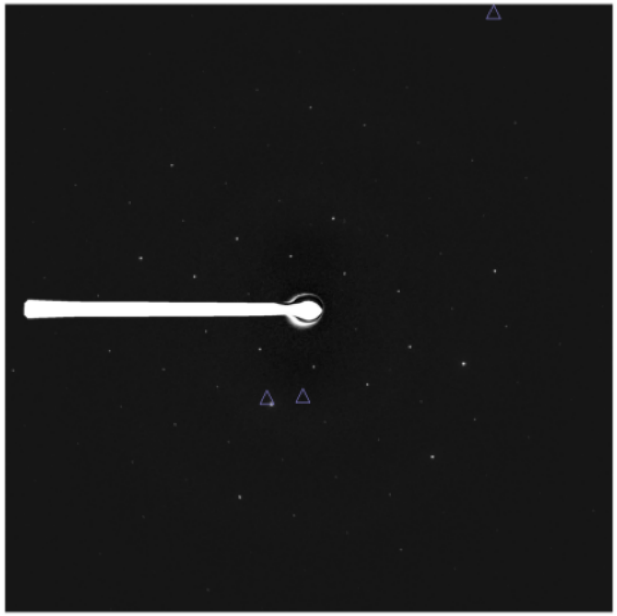
superxds / XDS background = BKGINIT x RATIO (log, same scale)



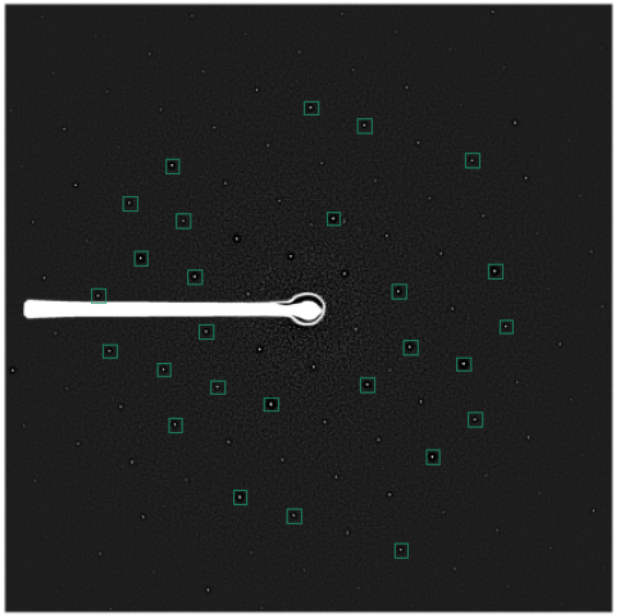
bgsub: frame - background, peaks (136)



superxds: frame - BKGINIT x RATIO, spots (3)

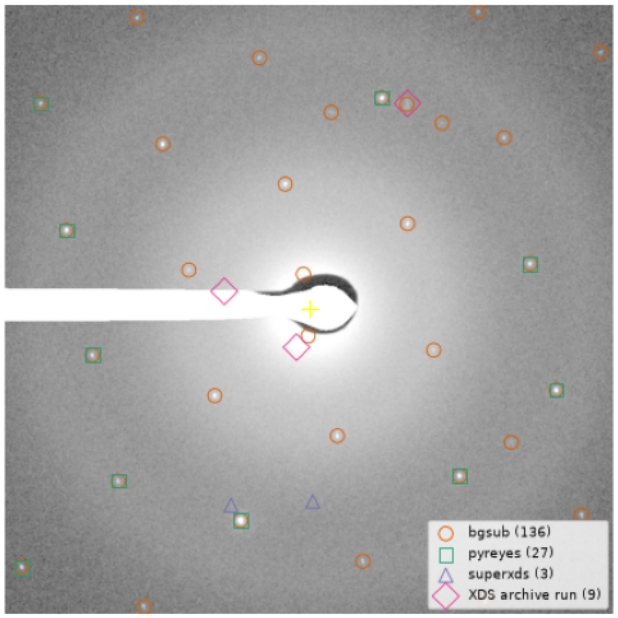


pyREyes: difference of Gaussians (no background), peaks (27)

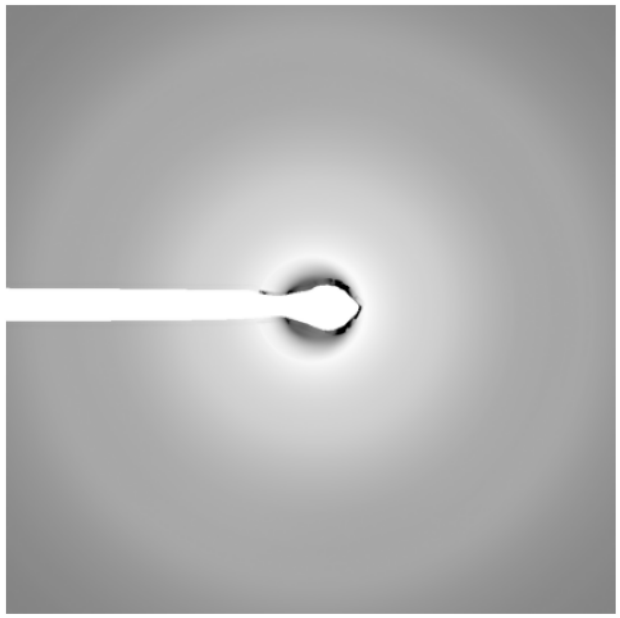


Arctica-CETA bin 2 (2048^2): rotation movie, 0.9 deg per frame, CL 960, CETA-D 4k binned 2 - central crop

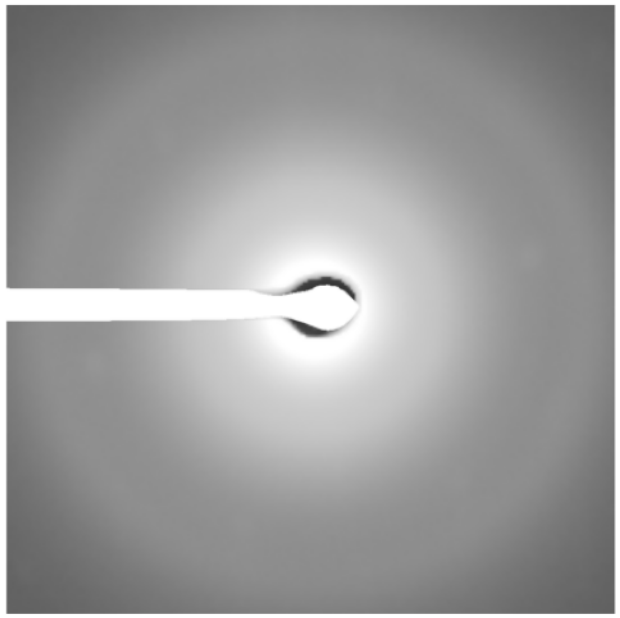
ceta_b2: original frame 56 (log), all methods' peaks



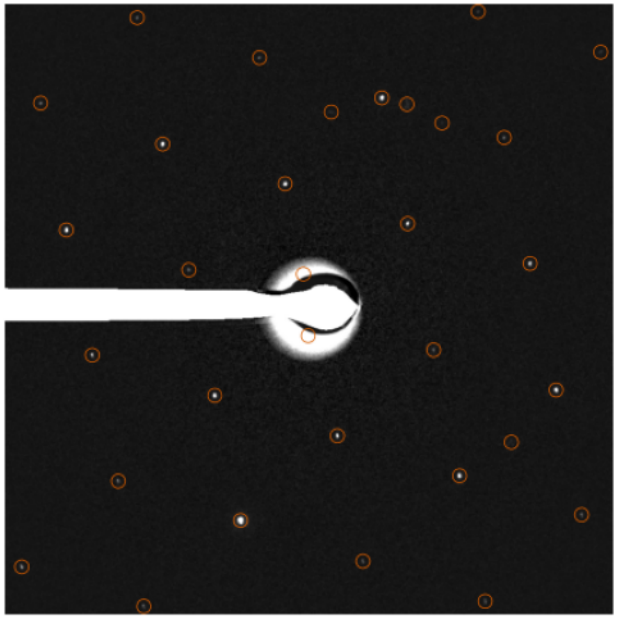
bgsub v2.1 background (log)



superxds / XDS background = BKGINIT x RATIO (log, same scale)



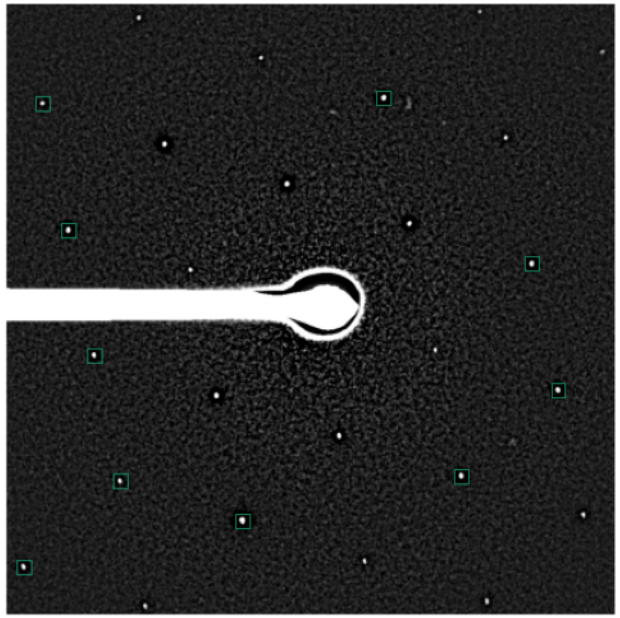
bgsub: frame - background, peaks (136)



superxds: frame - BKGINIT x RATIO, spots (3)

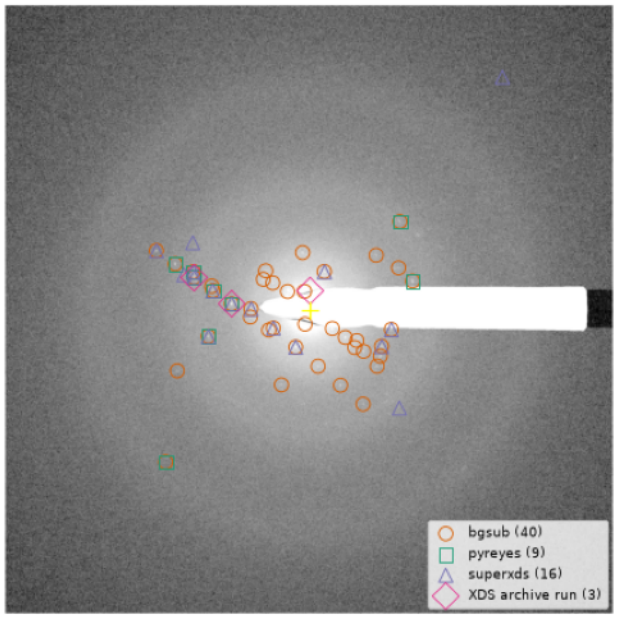


pyREyes: difference of Gaussians (no background), peaks (27)

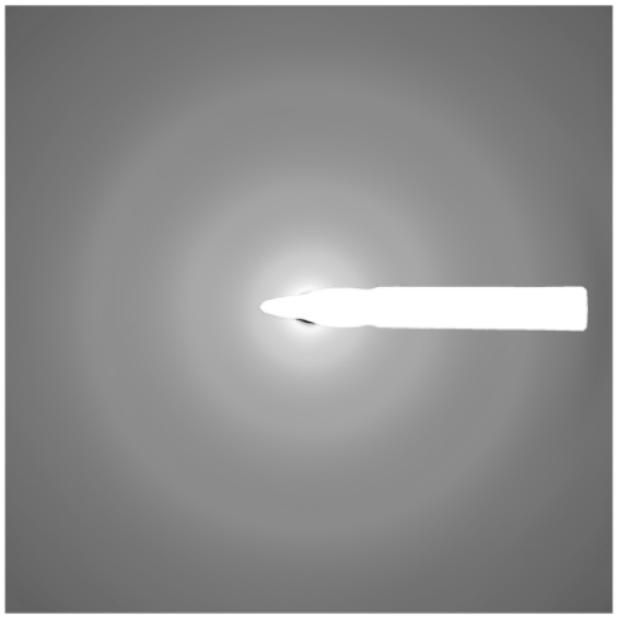


F30-TVIPS bin 2 (2048^2): rotation movie, 0.5 deg per frame, CL 1320 (300 kV), TVIPS 4k binned 2

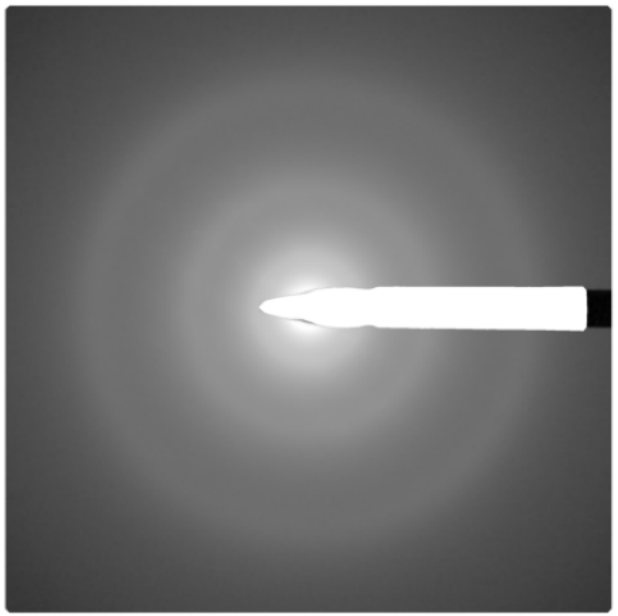
tvips_b2: original frame 50 (log), all methods' peaks



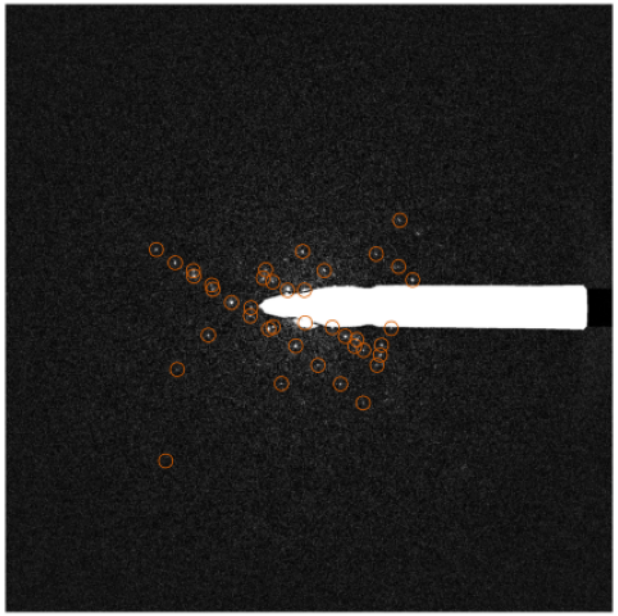
bgsub v2.1 background (log)



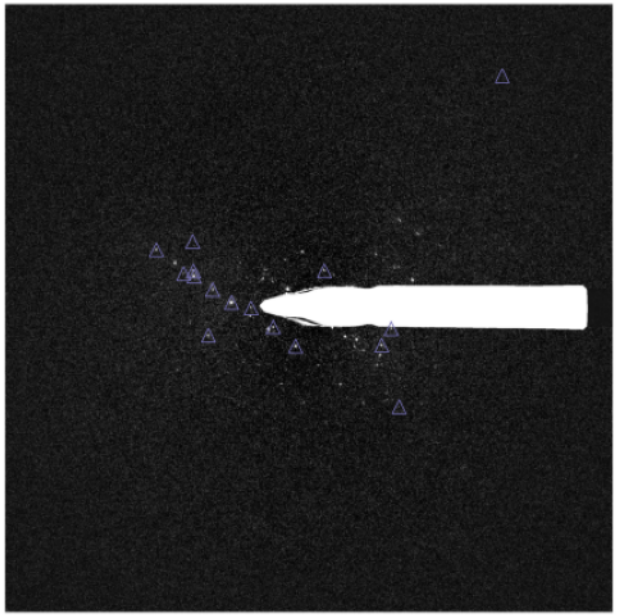
superxds / XDS background = BKGINIT x RATIO (log, same scale)



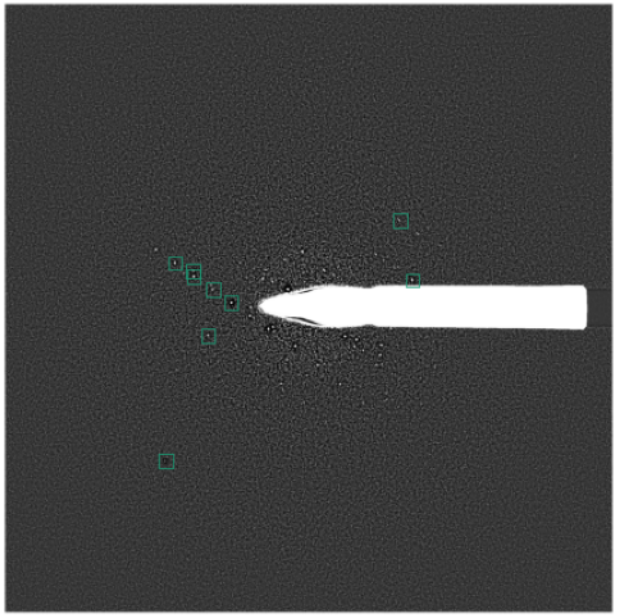
bgsub: frame - background, peaks (40)



superxds: frame - BKGINIT x RATIO, spots (16)

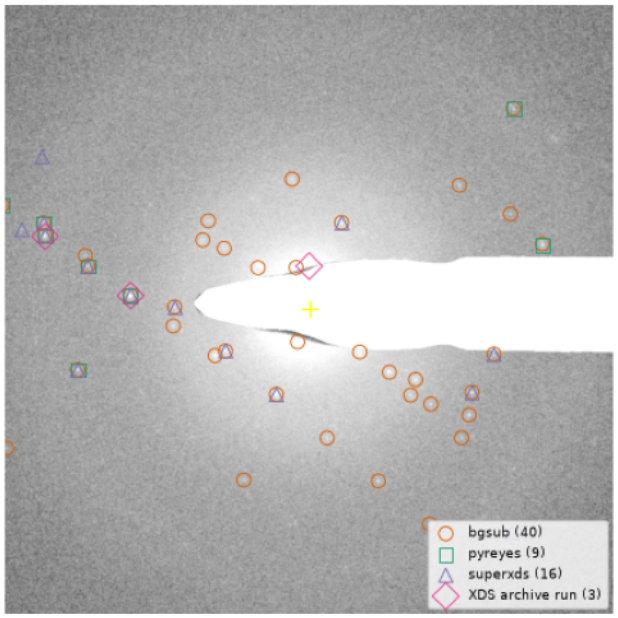


pyREyes: difference of Gaussians (no background), peaks (9)

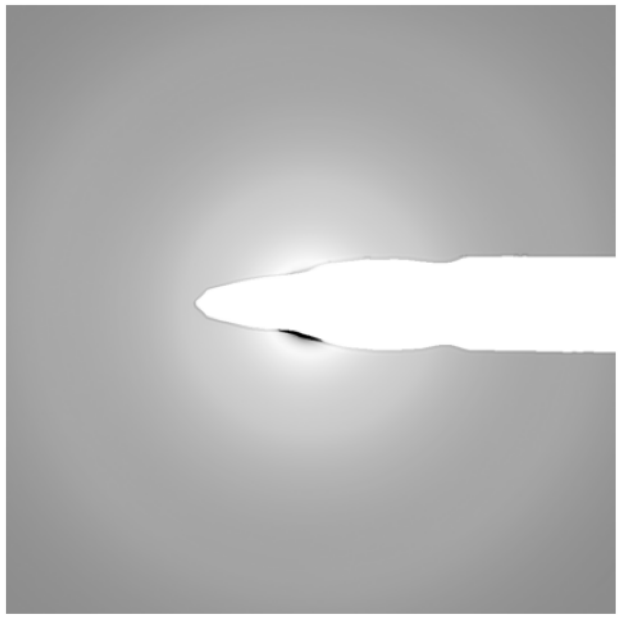


F30-TVIPS bin 2 (2048^2): rotation movie, 0.5 deg per frame, CL 1320 (300 kV), TVIPS 4k binned 2 - central crop

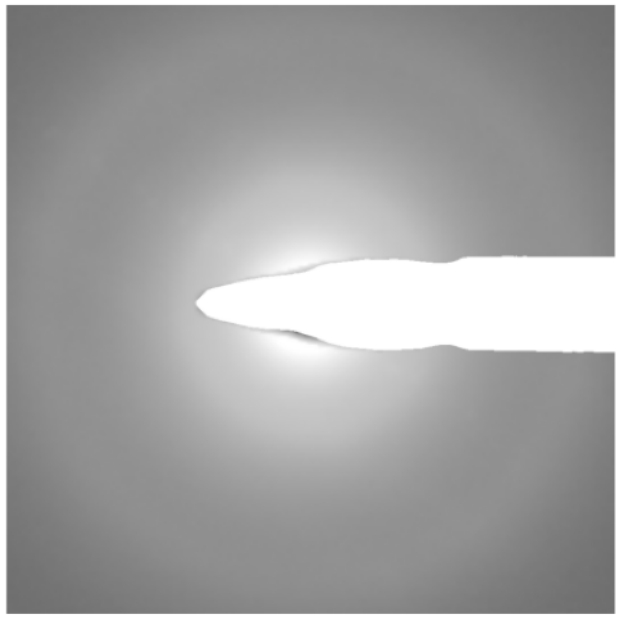
tvips_b2: original frame 50 (log), all methods' peaks



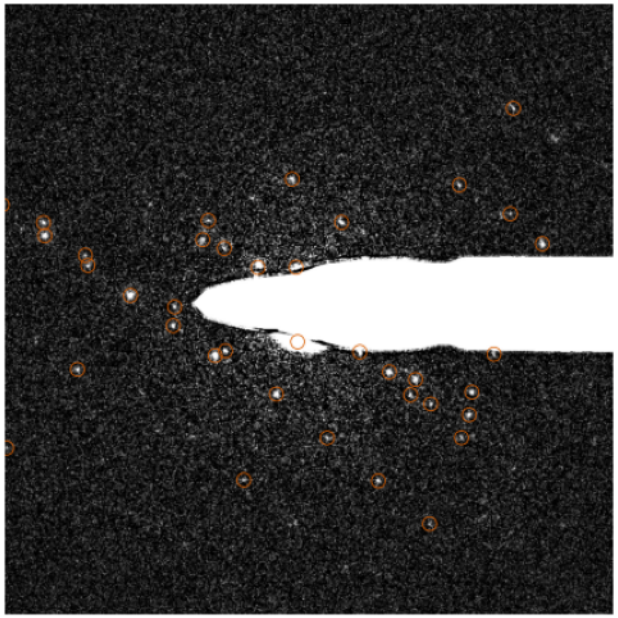
bgsub v2.1 background (log)



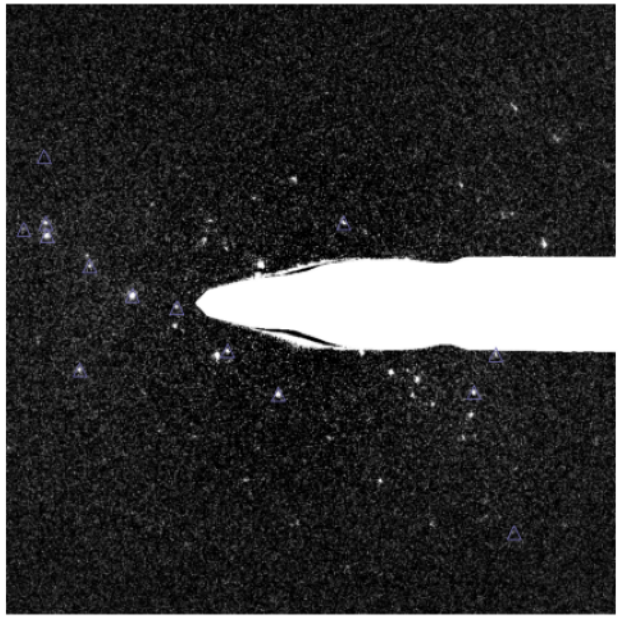
superxds / XDS background = BKGINIT x RATIO (log, same scale)



bgsub: frame - background, peaks (40)



superxds: frame - BKGINIT x RATIO, spots (16)



pyREyes: difference of Gaussians (no background), peaks (9)

